Aditya Kumar Singh

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Summary

Senior Software Engineer with dual expertise in **C++ System Programming** (5G-CRAN, DPDK) and **Java Application Development** (CPaaS, MRF). Proficient in optimizing Linux kernels for low-latency networking and architecting cloud-native solutions using Docker and Kubernetes. Experienced in handling high-concurrency traffic for Telecom and Finance sectors.

Education

•**Indian Institute of Technology (IIT - Hyderabad)** *2019 - 2022*
M.Tech in Computer Science Engineering | CGPA: 8.64 Hyderabad, India

•**University Institute of Engineering & Technology, CSJMU** *2014 - 2018*
B.Tech in Computer Science Engineering | CGPA: 7.89 Kanpur, U.P.

Experience

•**Radisys India Limited** *Mar 2023 - Present*
Senior Software Engineer - Cloud Engineering Bangalore

CPaaS & SIP Call Control (Java)

- **RESTful SIP APIs:** Engineered Java-based REST APIs and integrated the **Cipango SIP stack** via Servlets for real-time call signaling (INVITE, PRACK, BYE), ensuring robust carrier-grade telecom workflows.
- **SDP Parsing & Negotiation:** Developed advanced **Java utilities** to extract and validate media IP/ports via SDP, successfully diagnosing and resolving complex SIP interoperability and Via header compliance issues.
- **Code Modernization & DevOps:** Refactored legacy Java codebases for multi-tenant environments, improved error handling, and automated HTTP/HTTPS API endpoint deployments using **Shell scripts**.

5G-CRAN Platform (C/C++ & Cloud)

- **DU Feature Development:** Designed and developed multiple DU application features including Multi-FH and Multi-MH connections, enhancing network flexibility, scalability, and PM/FM Counters for Network metrics.
- **Kubernetes Orchestration & Golang:** Built a custom **Kubernetes controller in Go** to automate 5G-RAN profile creation, reducing deployment effort by **30%** compared to Helm while enforcing strict **CPU** and **HugePages** allocation.
- **System Performance Tuning:** Performed **Linux system-level optimization** for CU-DU modules (5G RAN) by tuning CPU governors and I/O paths; diagnosed **NUMA bottlenecks** using `pidstat`/`numastat`, cutting CPU requirement and system power usage by **15%**.
- **Advanced Debugging & Stability:** Performed root-cause analysis of **DU application failures** using **GDB and AddressSanitizer (ASan)**, proactively identifying and fixing complex memory corruption issues.
- **CI/CD & Code Quality:** Established a C++ unit testing framework achieving **90%+ code coverage**, effectively blocking regression bugs in the **CI/CD pipeline** before production release.

•**Dremio Software India Private Limited** *Aug 2022 - Feb 2023*
Software Engineer - Cloud Platform Hyderabad

- **Billing System Framework:** Designed and developed the core billing infrastructure for Dremio Cloud using **Java** and **MySQL**, ensuring accurate metering for high-volume data usage.
- **Marketplace Integration:** Integrated billing accounts with **AWS Marketplace** via TACKLE, enabling seamless automated payment processing.
- **Query Acceleration:** Optimized query performance for BI/SQL clients by implementing **Data Reflections**, reducing data retrieval latency by **30%** for critical analytical workloads.
- **Financial Integrity:** Engineered reconciliation logic to detect inconsistent replicas and automate credit reversals, ensuring financial accuracy across the cloud platform.

•**Indian Institute of Technology (IIT) Hyderabad** *Aug 2019 - Jul 2022*
Research Assistant - Intel 5GCRAN Hyderabad

- **Open Source Contribution:** Contributed to the **OpenAirInterface (OAI)** project for 5G-Cloud RAN, optimizing the architecture of L2/L3 RAN functions for improved energy efficiency.
- **Kernel Profiling:** Performed deep-dive energy profiling using **Intel VTune, perf, and GDB**, identifying and refactoring high-energy consumption paths in the kernel.
- **Testbed Optimization:** Optimized RAN performance on a real-time OAI testbed utilizing **USRP (B210/N300)** SDRs and programmable SIMs.
- **Automation:** Developed **Python** utilities for automated log parsing and massive data processing to visualize energy usage patterns across different O-RAN split options.

Technical Skills

- **Programming Languages**: C/C++, Golang, Java, MySQL, Python, Bash Scripting
- **DevOps & Tools**: Docker, Kubernetes, Git, Maven, CMake, LaTeX, IntelliJ IDEA
- **Performance & Debugging**: Intel VTune, Linux Perf, GDB, Wireshark, Address Sanitizer, pidstat, numastat
- **Domain & Simulation**: 5G-RAN, O-RAN, OpenAirInterface (OAI), NS-3, Low-Level Systems, CPaaS, SIP, MRF
- **Testing**: JUnit, Mockito, Google Test (GTest)

Projects

•**Master's Thesis: Cost-Benefit Analysis of O-RAN Deployment Scenarios** *Aug 2019 - Jul 2022*
IIT Hyderabad

- Conducted TCO Analysis (CAPEX & OPEX) for deploying different split options for CU-DU-RU placement in 5G Networks.
- Formulated cost-effective network deployment strategies for telecom operators based on regional population.

Research and Publications

- **Conference Paper**: U. Pawar, **Aditya Kumar Singh**, et al., "Understanding Energy Consumption of Cloud Radio Access Networks: an Experimental Study," *2020 IEEE 3rd 5G World Forum (5GWF)*, 2020. **[Link]**
- **Poster Presentation**: Shah Alam, **Aditya Kumar Singh**, "Design and Implementation of Image Cryptography using RSA Algorithm," *IEEE International Conference on Engineering Science & Advance Research*, 2018.

Certifications

- **Architecting with Google Kubernetes Engine: Foundations** - Coursera
- **Data Reflections** - Dremio Software
- **Dremio Fundamentals** - Dremio Software

Miscellaneous

•**Competitive Programming**: Active problem solver on LeetCode, HackerRank, and GeeksforGeeks, focusing on Data Structures & Algorithms (DSA) and System Design concepts.
•**Interests**: Cricket, Chess, Fitness Training